NETWEB TECHNOLOGIES

INDIA LIMITED

Empowering Compute, Network and Storage

DDR5 Memory Validation

Server 172.16.13.19 — Two Campaigns (256 GB + 128 GB)

Campaign A: 8 × Samsung 256 GB DDR5-6400 (2 TB)

Campaign B: 8 × Samsung 128 GB DDR5-6400 (1 TB)

On Tyrone MDI300 (dual Intel Xeon 6730P)

Prepared for:	Netweb Technologies India Ltd
Date issued:	28 May 2026
Test platform:	172.16.13.19 — same server, two memory configurations
Test windows:	Campaign A 08:25-09:50 UTC • Campaign B 10:13-10:58 UTC
Memory status:	**PASS (both) — 0 ECC errors, all 8 DIMMs detected, full 6400 MT/s**
Platform status:	**ACTION NEEDED — CPU locked at 500 MHz in BOTH campaigns (firmware)**

1. Executive Summary

Two memory configurations were validated back-to-back on the same server 172.16.13.19 (Tyrone MDI300, dual Intel Xeon 6730P “Granite Rapids”). **Campaign A**: 8 × Samsung 256 GB DDR5-6400 (2 TB). **Campaign B**: 8 × Samsung 128 GB DDR5-6400 (1 TB), after a physical swap. **The memory passes in both campaigns** — all 8 modules detected by BMC, BIOS and kernel, full rated 6400 MT/s, zero ECC errors. The same **platform fault persists in both**: every CPU core is locked at 500 MHz, which survived the reboot and DIMM swap — confirming it is independent of the memory and must be fixed before this server is benchmarked or deployed.

Metric	Campaign A (256GB/2TB)	Campaign B (128GB/1TB)	Verdict
Module P/N	M321RBJA0M22-CLPIL	M321RAJA0MB2-CCPWF	✓ Samsung
DIMM detect (BMC/BIOS/kernel)	8 / 8 / 8	8 / 8 / 8	✓ PASS
Configured speed	6400 MT/s (full)	6400 MT/s (full)	✓ PASS
30-min stress-ng verify	0 errors	0 errors	✓ PASS
15-min memtester locked	1920 GB (95.6%)	960 GB (95.9%)	✓ PASS
FAILURE/error in 64 logs	0	0	✓ PASS
EDAC CE/UE (16 mc)	0 / 0	0 / 0	✓ PASS
MCE events	0	0	✓ PASS
CPU core freq (under load)	500 MHz	500 MHz	! FAIL
1-thread memcpy (consequence)	2.31 GB/s	2.30 GB/s	! blocked

Bottom line: both DIMM sets are healthy and fully validated. The CPU frequency lock is reproduced identically in both campaigns — strong evidence it is a platform firmware issue, not a memory or workload effect. Bandwidth figures are throttled ≈10× and are not representative until the lock is cleared (see • 4 and • 5.2).

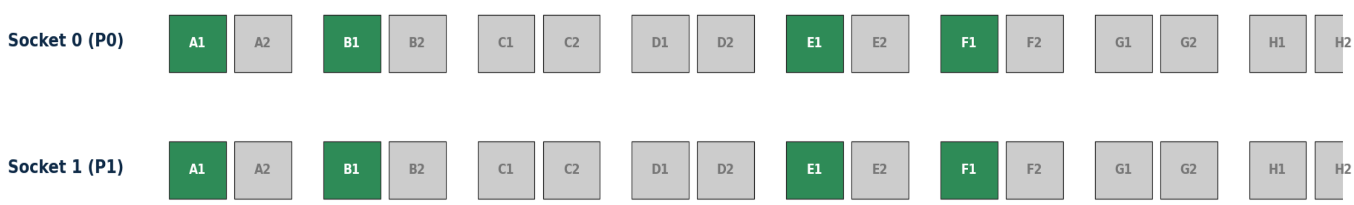
2. Test Environment & Detection

Attribute	Value (same server, both campaigns)
Server model	Tyrone MDI300 (dual-socket Xeon 6 server) at 172.16.13.19
BIOS / BMC	AMI ES418INW.M06 (16 Apr 2026) • BMC firmware 1.11
CPU	2 × Intel Xeon 6730P “Granite Rapids” (32C/64T each = 64C/128T)
CPU freq (rated)	Base 2500 MHz, max turbo 3800 MHz (AVX-512 + AMX, L3 576 MiB)
OS / kernel	Ubuntu 22.04.4 LTS / Linux 6.8.0-111-generic
EDAC	16 memory controllers (mc0-mc15), multi-bit ECC active
Slot population	8 of 32 slots: P0/P1 channels A, B, E, F (both campaigns)

2.1 Module configurations

Attribute	Campaign A	Campaign B
Module P/N	Samsung M321RBJA0M22-CLPIL	Samsung M321RAJA0MB2-CCPWF
Capacity / total	256 GB × 8 = 2 TB (2015 GiB)	128 GB × 8 = 1 TB (1007 GiB)
Rated / configured	6400 / 6400 MT/s	6400 / 6400 MT/s
Rank	Dual-rank	Dual-rank
ECC	Multi-bit ECC active	Multi-bit ECC active

2.2 DIMM population map (8-channel Granite Rapids, both campaigns)



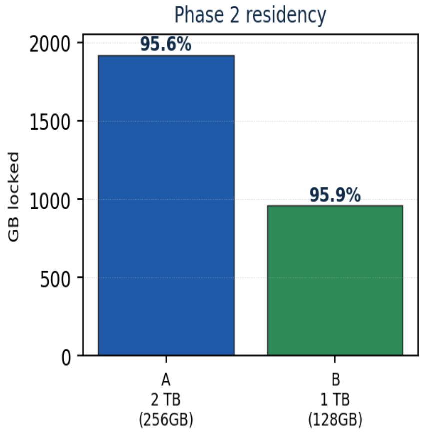
32 DIMM slots (2 sockets × 8 channels × 2 slots) • 8 populated = channels A,B,E,F slot 1, both sockets (1 DPC on 4-of-8 ch) • green = 256 GB (A) / 128 GB (B)

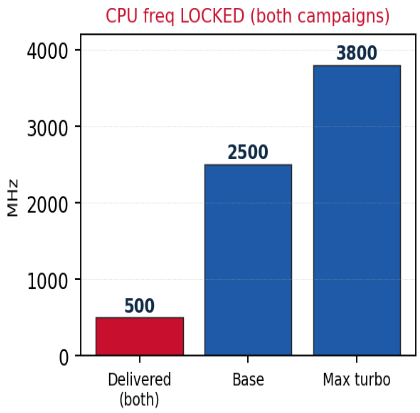
2.3 Detection chain — BMC + BIOS + kernel agree on 8 DIMMs (both campaigns)

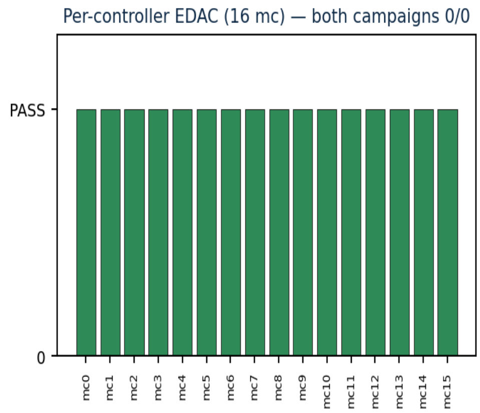
```
$ ipmitool sdr | grep -i DIM (Campaign B snapshot; Campaign A identical slots)
P0_DIM_A0 | 29 C | ok  P0_DIM_B0 | 28 C | ok  P0_DIM_E0 | 31 C | ok  P0_DIM_F0 | 32 C | ok
P1_DIM_A0 | 31 C | ok  P1_DIM_B0 | 32 C | ok  P1_DIM_E0 | 29 C | ok  P1_DIM_F0 | 28 C | ok
(remaining 24 slots: 'no reading | ns' = empty) — BMC sees exactly 8 populated DIMMs
```

BIOS DMI Type 17 lists 8 populated DIMM records in each campaign (correct P/N, 256/128 GB, 6400 MT/s). Kernel /proc/meminfo reports ≈2 TiB (A) / ≈1 TiB (B); EDAC enumerates all 16 controllers. No detection discrepancy at any layer in either campaign.

3. Memory Stability Results — Both Campaigns







Parameter	Campaign A — 256 GB / 2 TB	Campaign B — 128 GB / 1 TB
Phase 1 (30-min stress-ng)	08:25:18 → 08:55:24	10:13:33 → 10:43:38
Phase 1 working set	64 × 24 GB = 1536 GB	64 × 12 GB = 768 GB
Phase 1 result	0 errors; EDAC 0/0	0 errors; EDAC 0/0
Phase 2 (15-min memtester)	09:30:14 → 09:45:14	10:43:51 → 10:58:51
Phase 2 locked	64 × 30 GB = 1920 GB (95.6%)	64 × 15 GB = 960 GB (95.9%)
Phase 2 mlock()	succeeded on all 64	succeeded on all 64
Phase 2 FAILURE/error	0 across 64 logs	0 across 64 logs
EDAC delta (16 mc)	0 CE / 0 UE	0 CE / 0 UE
MCE events	0	0

Memory verdict: PASS in both campaigns. Zero ECC errors on all 16 controllers, zero memtester failures across 64 logs, mlock pinning ≥ 95 % of RAM for the full 15-minute window each time. Both the 256 GB and 128 GB Samsung DDR5-6400 module sets are electrically sound and correctly trained at full rated speed. (Per-pass throughput is low only because of the CPU lock in • 4.)

4. Platform Finding — CPU Locked at 500 MHz (both campaigns)

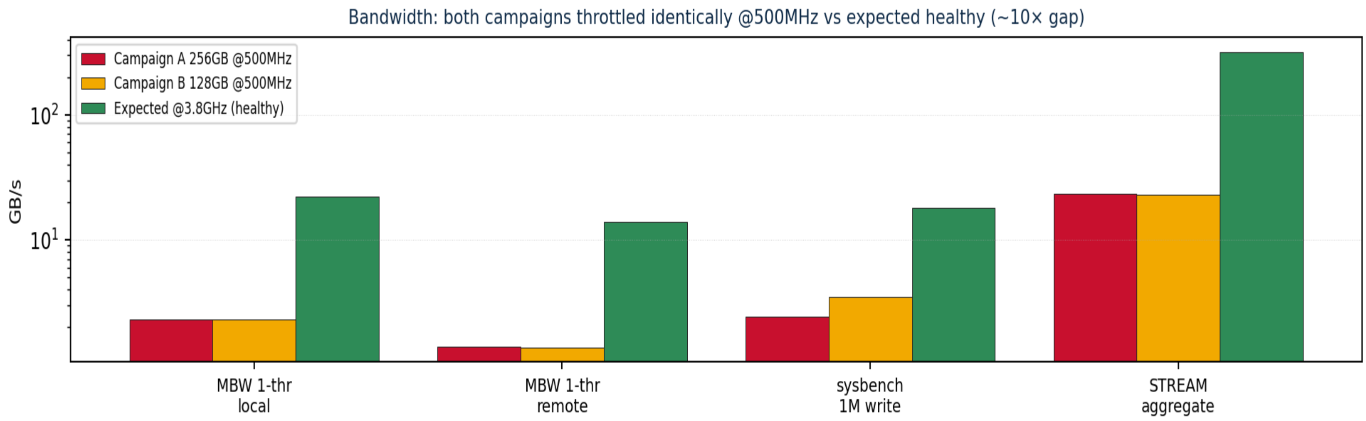
In both campaigns, every CPU core is pinned at **500 MHz** — one-fifth of the 2500 MHz base and one-seventh of 3800 MHz max turbo. The lock **persisted across the reboot and physical memory swap** between campaigns, proving it is a platform condition independent of the DIMMs. It is a firmware-level fault, not an OS or thermal condition.

Check	Observation (identical both campaigns)	Implication
turbostat Bzy_MHz (all-core load)	500 MHz on all 64 cores	Real delivered freq is 500 MHz
Governor / EPP / turbo	performance / performance / no_turbo=0 / max=100%	No OS-side restriction
HWP_REQUEST (0x774) & PERF_CTL (0x199)	ratio 0x26 (3800 MHz); wrmsr force ignored	OS requests 3.8 GHz; ignored
HWP_CAPABILITIES (0x771)	highest = 38 (3800 MHz)	Silicon claims 3.8 GHz capable
CoreTmp / PkgTmp / CoreThr	29-35 °C / 38 °C / 0	Not thermal throttling
RAPL pkg power vs limit	88-139 W vs 250 W cap	Not power-capped
CORE_PERF_LIMIT_REASONS (0x64F)	0	Core reports no perf-limit reason
dmesg	intel_uncore_frequency_tpmi: Unsupported minor version	Kernel 6.8 PM support incomplete for this silicon

Conclusion. The OS requests 3.8 GHz through every mechanism and the silicon reports it is capable, with no thermal or power limit active — yet cores deliver 500 MHz, identically in both campaigns. This is a **BIOS power-profile / config-TDP setting or early-Granite-Rapids pcode/microcode immaturity**, not resolvable from the OS and unrelated to the (healthy) memory.

5. Bandwidth Impact & Recommendations

5.1 Bandwidth under throttle — both campaigns (NOT representative)



Test	Campaign A @500MHz	Campaign B @500MHz	Expected @3.8GHz
MBW memcpy 1-thr local	2.31 GB/s	2.30 GB/s	20 - 24 GB/s
MBW memcpy 1-thr remote	1.40 GB/s	1.38 GB/s	12 - 15 GB/s
sysbench 1M rnd write 64thr	2.42 GB/s	3.49 GB/s	15 - 20 GB/s
STREAM aggregate	≈ 23 GB/s	≈ 23 GB/s	300 - 340 GB/s

Both campaigns produce nearly identical throttled bandwidth — further proof the bottleneck is the 500 MHz CPU lock, not the DIMMs. Recorded for completeness only; re-measure after the lock is cleared.

5.2 Recommendations

• **Memory: cleared for use.** Both the 256 GB and 128 GB Samsung DDR5-6400 sets pass all correctness and detection checks. No DIMM action required.
• **Resolve the 500 MHz CPU lock before benchmarking / deployment** (in order of likelihood):
(a) **BIOS power profile** — check Setup → Advanced → Power/Performance for a Max-Efficiency / fixed-low-ratio / config-TDP-low setting; set to Performance / max TDP.
(b) **Update BIOS + CPU microcode** — Granite Rapids is new silicon; the Apr-2026 BIOS may predate a pcode fix. Apply the latest Tyrone MDI300 BIOS + Intel microcode.
(c) **Check VR / power-delivery telemetry** — a VR fault can make the PUNIT self-limit to minimum frequency.
(d) **Newer kernel** (6.11+/HWE) — the “Unsupported minor version” uncore message indicates 6.8 lacks full PM support for this platform.
• **Re-run the bandwidth suite** once cores reach rated frequency — expect ≈ 300+ GB/s aggregate STREAM for this 4-of-8-channel DDR5-6400 layout.

5.3 Customer-facing wording

> *“On server 172.16.13.19 (Tyrone MDI300, dual Intel Xeon 6730P), Netweb engineering validated two memory configurations back-to-back — 8 × 256 GB DDR5-6400 (2 TB) and 8 × 128 GB DDR5-6400 (1 TB). Both passed full memory validation: detected by BMC, BIOS and kernel, trained at full 6400 MT/s, with zero ECC errors across a 30-minute pattern-verify pass and a 15-minute ≥95 %-RAM-resident burn each. Testing also identified a platform firmware issue locking all CPU cores at 500 MHz in both campaigns (vs 3.8 GHz rated); this is unrelated to the memory but must be resolved before performance benchmarking.”*